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Low-Cost Catamaran Tow Target (L-CATT) – Overview and Analysis

Overview

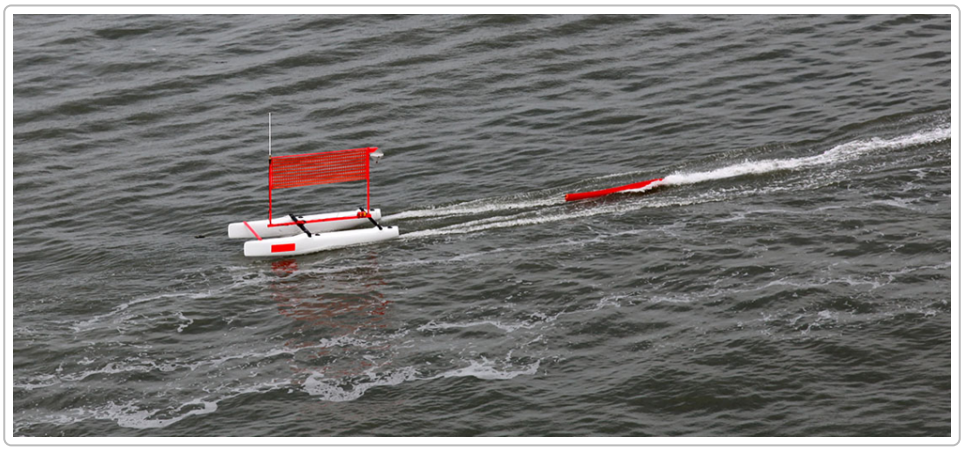

Figure: The Low-Cost Catamaran Tow Target (L-CATT) being towed at sea. It consists of two slender polythene hulls in a catamaran layout, used as a naval gunnery target.

The **L-CATT** (Low-Cost Catamaran Tow Target) is a **towed naval training target** designed to simulate small boat threats and serve as a live-fire practice target for naval weapons. Developed by QinetiQ Target Systems, it is built with **two roto-molded polythene hulls** in a catamaran configuration, giving a stable, buoyant platform that is **4.8 meters in length** [1]. The L-CATT is typically towed behind another vessel – ranging from manned ships to QinetiQ's own unmanned surface vehicles (USVs) like the Sprite II, Hammerhead, or Barracuda – and can function as an **expendable target for weapon testing and training** [1]. In practice, a towing craft will drag the L-CATT at operational speeds while gunners or missile systems engage the target, allowing realistic training without risking high-value assets. The use of a long tow line (often up to a few hundred meters) ensures the towing vessel stays at a safe distance during live fire [2].

Despite its simple design and low cost, the L-CATT can be outfitted with **optional payloads** to enhance threat representation. Depending on training needs, it can carry accessories like radar reflective screens, infrared emitters or smoke sources to mimic the signatures of different naval threats [3]. This flexibility means L-CATT can simulate a variety of targets – from small fast attack craft radar returns to the heat signature of a boat engine – by swapping payload kits. According to QinetiQ, the system has *"proven world-wide service"* with robust customer support, indicating that it has been adopted by many navies for their live-fire exercises [4]. It is designed to be **low-cost and expendable**, so if the target is destroyed during gunnery practice (which is often the intent), it is economically practical to replace [4]. The key features of L-CATT, as highlighted by its manufacturer, include:

- **Low-cost and expendable** – affordable enough to be used as a destroyable target [4].
- **Easily towable** – can be deployed and towed by a range of vessels (including QinetiQ's USVs or ordinary boats) without special modifications [1] [4].

- **Proven in service worldwide** – has been used by multiple customers globally with established support and reliability [5].

In summary, L-CATT is a simple yet effective tool for naval forces to conduct live-fire training. Its catamaran hull design gives it stability in the water, and its compatibility with high-speed towing platforms means it can present a moving target at realistic speeds. By being **sacrificial** (i.e. meant to be shot at and possibly destroyed), it allows crews to train with actual munitions – from machine guns and cannons up to missiles – thereby improving realism. The combination of low cost, ease of use, and adaptability has made L-CATT a **popular choice for gunnery practice and weapon system evaluation** in many countries [4]. For example, during a large Italian naval exercise in 2025, the navy employed L-CATT targets (alongside QinetiQ's Hammerhead USV targets) for live-fire drills, underscoring the system's value in providing realistic training scenarios [6].

Technical Specifications and Capabilities

In terms of construction and specifications, the L-CATT's defining characteristic is its **twin-hull (catamaran) design made of durable polythene plastic**. Each hull is roto-molded, a manufacturing process that produces a robust, hollow form; this likely makes the target inherently buoyant (often such targets are foam-filled or naturally buoyant so they remain afloat even after taking damage). The overall length of the target is **4.8 m (approximately 16 feet)**, which is about the size of a small boat or a personal watercraft [1]. The beam (width) is not explicitly stated in open sources, but being a catamaran, the two hulls are spaced to provide stability – typically one could expect a width on the order of 1–2 m. The height of any above-water structure is minimal, aside from any optional target aids (for instance, sometimes a small mast or an orange panel is mounted on the L-CATT to improve visibility or radar return). The photograph above shows an orange-red mesh panel on the L-CATT, which likely serves as a **radar reflector** or visual aiming reference for gunners. Such a panel can increase the radar cross-section (RCS) to ensure ship radars or missile seekers can lock onto the towed target if needed. QinetiQ literature indeed notes that optional payloads/panels can be added to *"effectively replicate various naval threats."* [3] This implies the L-CATT can be configured with, for example, a radar reflector (to simulate a larger vessel or to present a clear radar target) or perhaps an IR emitter (to simulate a boat's engine heat for infrared-guided weapons).

Being a **passive towed target**, the L-CATT has no engine or propulsion of its own. Its performance is therefore tied to the towing platform. The target is compatible with essentially any vessel capable of towing it; QinetiQ specifically mentions it works with all their unmanned surface vehicles and existing naval vessels [1]. In practice, users have to ensure the tow rope and L-CATT can handle the speeds and sea conditions of the exercise. While detailed speed ratings for L-CATT are not published in the snippet we have, we can infer capabilities from related systems. The **High-Speed Inflatable Towed Target (HSITT)**, which is an inflatable cousin of L-CATT, for instance, is designed for **towing up to 36+ knots** in calm seas, with decreasing max tow speeds as sea state rises (e.g. ~25 knots in Sea State 3) [7] [8]. The L-CATT, being a rigid-hull target, would likely have a similar high-end towing speed capability on the order of a few dozen knots (limited by the towing platform's power and sea conditions). Notably, QinetiQ's Hammerhead USV – one of the platforms that can tow L-CATT – can itself travel **35+ knots in Sea State 3** [9], so the L-CATT is presumably engineered to be stable at those speeds when towed. Its catamaran form and low profile help it **skim over the water with reduced drag and stable tracking** behind the tow vessel. Operators report that the HSITT (inflatable target) uses a patented skeg (keel) design to prevent rolling or instability at high speed [7]; the L-CATT likely relies on its twin-hull geometry for similar stability in tow.

Another important specification is weight: while we don't have an exact figure for L-CATT's weight, it is meant to be man-portable or easily handled on deck. Comparable targets (like inflatable HSITT Mk II)

weigh around 68 kg (150 lbs) dry [10], though a rigid plastic target might weigh a bit more. Even so, the L-CATT would be in the low hundreds of kilograms at most, making it feasible for a small team to deploy it using a ship's crane or even manhandle it off a slip if needed. The **tow line length** can vary; QinetiQ's Sprite II USV (a 4.2 m RHIB) can be equipped with a **500 m tow reel** specifically for deploying and recovering towed targets like L-CATT [2]. In actual training, using a few hundred meters of tow line is common to ensure safety – this distance keeps the target far enough from the towing ship so that misses or near-misses do not endanger the towing platform, and also to prevent muzzle blast or missile exhaust from affecting the towing vessel.

Durability: L-CATT is described as *"low-cost and expendable,"* which suggests it is **expected to absorb live fire** and potentially be destroyed [5]. Its polythene hulls can likely withstand splinters or small-caliber hits to some degree (polythene is tough and somewhat pliable), but direct hits from medium or large naval weapons will puncture or break the hulls. Being expendable, the target does not need to survive the engagement – in fact, a successful gunnery exercise might result in a sunk or heavily damaged L-CATT. If it isn't destroyed, crews can recover it, patch or repair minor damage, and reuse it. (For context, the inflatable HSITT target can survive multiple 20–30mm cannon hits due to its multi-chamber design, and can be quickly patched and re-inflated for reuse [10]. A rigid L-CATT might not be as forgiving with large holes, but minor hits could be tolerable.) QinetiQ emphasizes **ease of repair and support**, and since L-CATT has a simple modular build (two hulls and cross-beams), repairing or replacing a hull section could be relatively straightforward if the customer has spare parts.

Visibility and Tracking: In training exercises, it's crucial that both the trainees and the test instrumentation can track the target. The L-CATT's bright coloring (often painted high-visibility orange/red and white) makes it stand out against the sea for visual tracking. The optional enhancements like radar reflectors (e.g., corner reflectors or radar-enhancing mesh panels) give a radar signature akin to small craft – QinetiQ offers a passive RF reflector of 20 m^2 RCS as a payload option for targets [11] [10], which likely can be used with L-CATT as well. There may also be the possibility to mount a small **smoke generator or pop-up flare** to simulate a burning vessel or to mark the target's location. The target can also tow something itself – for example, sometimes a floating smoke marker or smaller decoy can be trailed behind to simulate a wake or threat. However, by itself L-CATT is primarily a simple platform – any such advanced functions would come from add-on kits.

In summary, the technical profile of L-CATT is one of **simplicity and robustness**: a 4.8 m twin-hull plastic boat that can be easily towed and equipped with augmentations to suit the training mission. It does not have onboard electronics mentioned (no engine, likely no active telemetry of its own), but the towing vessel or range control might track it on radar or via visual/IR means. Its value is in presenting gunners or missile crews with a **realistic moving target** over water – one that has the physical characteristics (size, speed, wake) of a small hostile craft or incoming threat, thereby providing more realistic training than firing at static targets or virtual simulations. The minimal technical complexity of L-CATT also translates to high reliability – there are few things that can go wrong during an exercise (no engine to stall, no remote control system to fail). As long as it's properly attached to the towline, it will do its job of following along as a target. This "dumb" simplicity is a deliberate design choice for an expendable target.

Comparisons with Similar Target Systems

L-CATT belongs to the family of **naval target systems** and has a few notable counterparts and alternatives, both within QinetiQ's product line and from other defense companies. Below we compare L-CATT with some similar systems:

- **High-Speed Inflatable Towed Targets (HSITT)** – QinetiQ's HSITT MkII is an inflatable tubular target that serves a similar purpose as L-CATT (being towed for gunnery practice) but with a different design approach. Instead of rigid hulls, HSITT is essentially an 7.3 m long inflatable "sleeve" or bladder made of reinforced PVC, divided into multiple air chambers [12] [13]. This gives it the advantage of being very lightweight (only ~68 kg) and **easily stowable**, as it can be deflated and rolled up for storage [10]. The HSITT was *"primarily designed to be towed by the Barracuda USV-T at high speeds"* [14] and features a patented water **skeg (keel) for stability**, preventing it from rolling or submarining even at **36+ knots tow speed** [7] [15]. In contrast, L-CATT, with its solid catamaran form, might be a bit bulkier to transport (it occupies more volume on deck) and possibly cannot pack down like an inflatable. However, both are **low-cost, expendable targets** and indeed share similar roles. HSITT is noted as being *"in service with over 10 militaries worldwide"* and *"recognized as the best naval inflatable targets in the world"* [8] – indicating a high degree of confidence and widespread use. L-CATT's "proven worldwide service" claim suggests it too has multiple users, though likely the HSITT (and its earlier versions) preceded L-CATT in many inventories. One key difference is durability: HSITT can survive multiple hits because bullet holes in air chambers only slowly deflate the target (and it has redundant chambers to stay afloat) [12]. L-CATT's plastic hulls might crack or flood when hit, so in terms of **survivability across an entire exercise**, the HSITT might last a bit longer under fire. On the other hand, if not hit, L-CATT could potentially be reused indefinitely, whereas inflatables can suffer wear or punctures over time as well. Both systems can be **anchored** to serve as a stationary target too, if needed (HSITT can be sea-anchored and used as a drifting target) [14], and presumably L-CATT could likewise be moored in place for target practice in a fixed location (QinetiQ's services even list a "marine static target L-CATT" as a hardware option [16]).

- **Towed Target Rafts/Sleds vs L-CATT:** Before modern systems like L-CATT, navies often used simple rafts or sleds as gunnery targets. For instance, the U.S. Navy had the **Williams Sled** and the **HARM Barge** – essentially floating platforms with structures to present a radar/visual target for guns or anti-radiation missiles [17] [18]. These were larger and heavier, and required larger vessels to tow. The L-CATT can be seen as a more advanced replacement for those: it's smaller and more maneuverable, enabling use in *Fast Inshore Attack Craft* (FIAC) scenarios. In fact, the U.S. Navy's development of a **Low-Cost Modular Target (LCMT)** parallels the L-CATT concept. The LCMT was envisioned as a **single modular towed target** that could present different profiles (with mission kits for gunnery, anti-ship missile, Hellfire missile, and HARM missile training) [17]. The goal was a *"lower cost, increased survivability"* target to replace older ones (like the inflatable ISTT, Williams sled, etc.) [19] [20]. This description is strikingly similar to L-CATT's purpose – a versatile, cheap target that can take on different roles by swapping payloads. A slide from a 2009 Naval Air Warfare Center presentation even shows an image of **"LCTT" (Low Cost Tow Target)** which looks like a small catamaran with an orange sail, not unlike the L-CATT's design [21] [17]. It's likely that QinetiQ's L-CATT (or something very similar) was offered to or adopted by some navies pursuing that concept. Thus, in a broad sense, L-CATT fits into the category of **"low-cost modular targets"** that are trending in modern naval training – replacing old heavy target barges with something agile and cheap enough to be expendable.

- **Self-Propelled Unmanned Surface Targets:** Instead of towing a target, an alternative is to use a **remote-controlled target boat** that maneuvers on its own. QinetiQ offers the **Hammerhead**

Mk II USV-T (5 m long, 135 hp engine) and the **Sprite II** (4.2–7.5 m RHIB-based USV) as such targets [9] [22]. These craft can simulate the actual tactics of a fast attack boat – e.g., weaving, swarm attacks – under remote or pre-programmed control. They can also serve as tow platforms for L-CATT or HSITT, essentially acting as the "driver" for towed targets when a manned vessel is not used [2] [23]. In comparison to L-CATT: the **Hammerhead/Sprite** approach gives a more complex and realistic target (since the vessel itself is the target and can maneuver), but they are **much more expensive assets** (Hammerhead is a powered drone boat with sophisticated control systems) and typically are **not meant to be destroyed** in training – they often tow an expendable target like L-CATT to take the hits instead [23]. L-CATT complements these by being the "disposable" element. For example, a Hammerhead USV can conduct high-speed evasive maneuvers while towing an L-CATT 500m behind it – the navy gunners will aim for the L-CATT, allowing the Hammerhead to escape damage if all goes well. If a missile or shell accidentally hits the Hammerhead, that's a far more costly loss than losing the L-CATT. Thus, even in the era of unmanned target boats, towed targets like L-CATT remain very relevant. Competitors to QinetiQ in this arena include companies like **L3Harris**, which produces the **C-Target series** of unmanned surface targets. The L3Harris **C-Target 13**, for instance, is a 13-meter target drone boat that can operate independently or **serve as a tow boat for inflatable targets or other small craft** [24]. This is analogous to QinetiQ's Barracuda (a larger high-performance USV) or other high-speed drone boats. However, not all navies can afford a fleet of drone boats for target practice – which is why a simple towed target (pulled perhaps by any available patrol boat or even an aircraft tug boat) is attractive. L-CATT can also be towed by a ship's helicopter in theory (though that is more commonly done with aerial towed gunnery targets for aircraft training). In naval gunnery, typically a small boat or unmanned RHIB tows the target at a safe distance and the warship practices with its main gun or CIWS. In that sense, L-CATT's direct alternatives are things like the old **"killer tomato" inflatable targets** – bright orange inflatable cubes/spheres used by the US Navy for decades – and the more modern inflatable or modular targets. Compared to a static inflatable balloon, the L-CATT gives a more boat-like behavior (skimming on water, creating a wake, etc.), which can be important for training sensors and gunners to recognize and track real threats.

- **Cost Comparison:** While exact pricing is not public, the term *"low-cost"* in L-CATT's name is relative – in the defense context, it implies on the order of tens of thousands of dollars (or less) per unit, as opposed to hundreds of thousands or millions for more complex targets. An expendable aerial target drone (like a Banshee) might cost $50k-$100k+ each; a large unmanned target boat can cost even more. An inflatable naval target might cost a few tens of thousands. L-CATT, being very simple, likely aims to be at the lower end of these costs. Its value proposition is that you **don't need a highly specialized towing platform** (any suitable vessel will do) and you can reuse it until it's destroyed, at which point replacing it won't break the budget. By comparison, a sophisticated USV target could cost significantly more and requires maintenance, trained operators, etc. So for navies or units that want to conduct live-fire training regularly, having a stock of L-CATTs (and/or HSITTs) is a cost-effective solution. They might reserve the fancy unmanned boats for more complex scenario training, and use the basic towed targets for everyday gunnery practice and weapon test calibration.

In essence, L-CATT fills a niche of **simple, modular naval target** that bridges the gap between ultra-basic targets (like floating barrels or balloons) and expensive autonomous target drones. Its peers include inflatable towed targets, small target drones, and older modular target platforms – and in many ways, L-CATT either augments or is poised to replace those in modern training exercises.

User Feedback and Operational Use

Because L-CATT is a military training device, formal "user reviews" are usually in the form of after-action reports or procurement feedback rather than public reviews. However, the **operational track record** of L-CATT provides insight into user satisfaction. QinetiQ's statement that L-CATT has *"proven world-wide service"* with extensive customer support suggests that multiple navies have successfully integrated it into their training programs [4]. In practice, a good training target is one that is reliable, easy to deploy, and adds realism to exercises – and the recurring inclusion of L-CATT in exercises indicates it meets those criteria.

Several navies have utilized QinetiQ's surface targets (including L-CATT) in live-fire drills. For example, the **Royal Canadian Navy (RCN)**, under a long-term contract with QinetiQ Target Systems, regularly conducts exercises where a variety of targets are presented. In the 2025 **Cutlass Fury** exercise (a Canadian-led multinational naval exercise), QinetiQ provided a suite of targets including aerial drones and Hammerhead USV targets [25]. While that press release highlights Hammerheads and aerial targets, the inclusion of towable targets like L-CATT in such inventories is common for close-range gunnery practice (for naval gun crews to fire at something like a small boat representation). The fact that QinetiQ's team operated **up to 40 targets simultaneously in swarm scenarios** during training indicates how these targets are seen as **force-multipliers for realistic drills** [26] – multiple L-CATTs could be towed to simulate a swarm of attacking craft, for instance. The RCN's continued renewal of contracts with QinetiQ implies satisfaction with the performance and training value of these targets.

A concrete example of user employment comes from the Italian Navy during **Exercise Mare Aperto 2025** (one of Italy's largest naval drills). They introduced new target systems to increase realism, including QinetiQ's aerial targets (Banshee drones) and surface targets. The Italian Navy specifically used *"LCATT and new Hammerhead MkII"* targets in live firing exercises during this drill [21]. Italian FREMM frigates and destroyers fired their 127 mm and 76 mm guns at these targets, including night shoots, to simulate engaging fast-moving threats [6]. The exercise report noted that this *"introduction of new naval and air targets… increased the realism of training"* [6]. This is an implicit endorsement: from the user's perspective, the L-CATT (towed by Hammerhead USVs in this case) provided a realistic challenge that improved training outcomes. The fact that it was used in a high-profile exercise with participants from many NATO countries suggests confidence in the system's safety and efficacy – it performed without issues significant enough to mention, even under the stress of a large exercise.

Users often appreciate the **simplicity** of L-CATT. Setting up the target is straightforward – crews attach the tow line bridle to the towing vessel, deploy the target into the water, and it's ready to go. There are no engines to fuel or electronics to configure on the target itself. A common piece of feedback for such systems is that they **free up operational flexibility**: because the target is so low-cost, commanders can plan live-fire events with less worry about target expense. If a navy wants to practice with expensive missiles (like anti-ship missiles or CIWS missiles), losing a target drone boat every time would be prohibitive. But using an L-CATT (perhaps augmented with a radar enhancer to register on missile seekers) means they can actually fire a live missile at a real moving target. If the missile hits, the target may be destroyed – which actually provides valuable data and confidence in the weapon – and the cost is limited to the expendable target, not a whole drone craft. This was highlighted by QinetiQ's marketing: L-CATT is *"low-cost and expendable"* by design [4], which is exactly what users need for live-fire testing and **weapon evaluation**. Many weapon system tests (for ship guns, or new naval missiles) have used such targets to prove out systems. For instance, a new auto-cannon or a guided projectile might be tested by firing at an L-CATT to verify accuracy and lethality. The users (test officers, gunnery instructors) have noted that having a **real target with realistic movement** greatly aids in evaluating tracking systems and operator performance. A moving target like L-CATT forces the gun crews to

account for target speed and sea conditions, making the training much more valuable than shooting at a static towed sleeve or a simple floating cone.

In terms of any criticisms or challenges: one consideration users mention is **sea state limits**. A lightweight target like L-CATT can be affected by very rough seas. While it is stable in moderate chop (and the catamaran design helps), in very high sea states the target might become less visible (hidden by waves) or could even flip if hit by a large wave at high towing speed. However, most live-fire training is scheduled in reasonable weather conditions, and L-CATT would typically be used in sea states 1–3 comfortably. Another point is tracking at long range: because the L-CATT is relatively small (4.8 m long), at long distances it's a tiny radar contact and almost a speck optically. That's why users often employ the aforementioned enhancements (radar reflectors, bigger flags, or even attaching a larger floating balloon above it) for long-range missile shoots. Feedback from exercises suggests that when properly equipped, L-CATT can reliably be engaged by ship sensors and weapons. For example, if a ship wants to test its radar-guided gun at 5+ km, the L-CATT might tow an corner-reflector or have a radar enhancement so that the fire control radar sees it well. Users have also noted the importance of a good **tow mechanism** – the tow rope and bridle need to be strong and preferably neutrally buoyant (so it doesn't sink and snag). QinetiQ's provision of an electric winch with 500 m line on Sprite II and similar shows they addressed this, making deployment and recovery smoother [2]. In exercises where multiple targets are used, some coordination is needed to avoid tangling lines or confusion between different targets – but that's a general exercise planning issue, not specific to L-CATT.

Overall, **user sentiment toward L-CATT is positive** given its continued and expanding use. It's seen as *"does the job"* for a fraction of the cost of high-end targets. Navies that have integrated it benefit from more frequent live-fire training since the consumable cost per exercise is lower. QinetiQ's support (often they send field service teams to help run the targets during major exercises [27] [28]) is another aspect users appreciate – they can rely on experts to maintain and operate the targets, allowing the navy personnel to focus on the training itself. In summary, L-CATT and similar tow targets have become **indispensable tools for naval training**, receiving praise for their **affordability, reliability, and realism**. The combination of user feedback and exercise outcomes suggests that the L-CATT effectively meets the training requirements it was designed for, providing naval crews with a target they can shoot, hit, and learn from with confidence.

Sources: The technical details and features of L-CATT are based on QinetiQ's official description [1] [3] and related product literature. Comparisons draw on QinetiQ's data for the HSITT inflatable target [12] [7] and on industry information about analogous systems like L3Harris C-Target and U.S. Navy target programs [24] [17]. Examples of operational use are documented in naval exercise reports and press releases, such as the Italian Navy's Mare Aperto exercise in 2025 where L-CATT was deployed [6]. These sources collectively illustrate the L-CATT's role, performance, and reception among its users.

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